Falak Tulsi

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Education

University of California, Riverside — **Riverside, California**
M.S. in Computer Science (GPA: 3.81/4.0) — *Expected December 2026*
B.S. in Computer Science (GPA: 3.83/4.0) — *Conferred June 2025*

Relevant Coursework: Software Construction, Data Structures and Algorithms, Artificial Intelligence, Natural Language Processing, Computer Architecture, GPU Architecture & Parallel Programming, Advanced Operating Systems, Information Security, Compiler Design, Embedded Systems

Technical Skills

Languages: Python, C++, Rust, Go, TypeScript, SQL, C
Frameworks & AI Tooling: Model Context Protocol (MCP), PyTorch, TensorFlow, OpenCV, Natural Language Processing (NLP), Hugging Face, React, FastAPI, Node.js
Systems: Linux/Unix, Docker, Datadog, Kubernetes, PostgreSQL, Git, CMake, GTest, Valgrind

Experience

Juniper Square — **San Francisco, CA (Remote)**
AI Engineer Intern — *June 2026 – August 2026*

- Built a deterministic AI telemetry pipeline for Claude Code and Cursor, replacing unreliable LLM self-reporting with IDE lifecycle hooks, eliminating silent event drops for **30+ skills** and feeding executive Datadog dashboards for **70+ engineers**
- Architected an autonomous self-documenting agent using **MCP (Model Context Protocol)** to index repository capabilities on every PR, conducting **Rapid Risk Assessments** (RRA) and **OSS security reviews**
- Designed a knowledge distillation engine via the Confluence REST API, building a PoC that automatically captured and structured **~10 real cross-team technical learnings** from raw developer session logs
- Co-developed an extensible AI QA platform unifying test case generation, gap analysis, and Playwright workflow execution to establish a standardized AI baseline for enterprise QA onboarding

Hewlett-Packard Inc. — **Palo Alto, California**
Software Engineer Intern (AI Systems) — *June 2025 – December 2025*

- Designed and deployed a local **LLM inference platform** using Lemonade-Server and Open WebUI, reducing inference latency **60%** via GPU acceleration
- Built automated **Python benchmarking infrastructure** to evaluate model performance across hardware configurations, increasing experimentation throughput $3\times Implementedtelemetryandloggingpipelinestotrackperformanceacrossinferenceworkloadsandhardwareenvironments$
- Analyzed architecture trade-offs between **cloud and on-device inference**, justifying business objectives of cost-efficiency and data sovereignty to executive leadership

Projects

Financial Search Engine | *Python, Scrapy, PyLucene, FAISS* — **UC Riverside Graduate Team Project (CS 242)**

- Built a **Scrapy-based distributed crawler** collecting **500MB+** of financial news and market data across **4 worker nodes**, implementing URL normalization, duplicate detection, and frontier coordination to prevent cross-site re-crawling
- Designed a **Lucene index** with custom analyzers for financial entities (**500+ S&P ticker symbols**, company names, headlines) enabling structured filtering across **8 indexed fields**
- Implemented **semantic retrieval** by generating **768-dimensional BERT embeddings** for **512-token document passages** and indexing them with **FAISS** to compare dense retrieval against Lucene keyword ranking

xv6 Kernel Development | *C, RISC-V, Operating Systems* — **UC Riverside Graduate Coursework (CS 202)**

- Implemented **copy-on-write fork** in the xv6 kernel to eliminate unnecessary memory duplication during process creation
- Engineered **Read-Copy-Update (RCU)** synchronization primitives enabling scalable multi-core read concurrency

NoFlame (SBHacks XI Winner) | *Python, Flask, TensorFlow, ESP32* — **github.com/HackSprout/NoFlame-repo**
1st Place Winner — **devpost.com/software/noflame**

- Built a wildfire detection system combining **computer vision**, machine learning, and IoT sensors, achieving **92% CNN accuracy** with real-time OpenCV monitoring
- Developed a Flask backend integrating geolocation, weather, and sensor telemetry to trigger automated alerts through ESP32 devices